6.1 Lagrange's equations with constraint

<Two types of constraints>

① Holonomic	② Pfaffian constraint
• $h(q) = 0$ • $\frac{\partial h}{\partial q}\dot{q} = 0 \Rightarrow$ Pfaffian • Actual definition is the constraints that restricts q to ~~the~~ a smooth hypersurface of Q.	• $A(q)\dot{q} = 0$ • If exact, it is holonomic • Pfaffian constraint restricts velocity instead of q.
(ex) Pendulum : $x^2 + y^2 = l^2$	Grasping : $J_h(\theta, x_o)\dot{\theta} = G^T(\theta, x_o) V_p$
• Basically lowers the dimension of $Q \Rightarrow$ lowers the degree of freedom	• Adds an additional force that enforces constraint ($A^T\lambda$). For example, normal force and static friction force in grasp th • These additional force do no w : d'Alembert's principle.

※ Why $A^T\lambda$?

Additional force should have no work when velocity obeys $Av =$

$\Rightarrow \forall v \in N(A)$, $P = F^T v = 0 \Rightarrow F \in N(A)^{\perp} = R(A^T) \Rightarrow F = A^T\lambda$.

<Lagrange multipliers>

$L(q, \dot{q})$: Lagrangian for the unconstrained system.

$A(q)\dot{q} = 0$: constraints. Υ : Additional (external) force

$\Rightarrow$ Euler-Lagrange equation : $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} + A^T(q)\lambda(q, \dot{q}) - \Upsilon = 0$.

If $L(q,\dot q) = \frac{1}{2}\dot q^T M \dot q - V(q)$, then earlier chapter's result gives

$M\ddot q + C\dot q + N + A^T\lambda = F$. Use $A\dot q = 0 \Rightarrow A\ddot q + \dot A \dot q = 0$ to remove $\ddot q$

$\Rightarrow (AM^{-1}A^T)\lambda = AM^{-1}(F - C\dot q - N) + \dot A \dot q$. So we get λ in terms of $q, \dot q$.

(Actually it is a Lagrange multiplier method)

(Example) Pendulum.

$Q = \mathbb{R}^2$. $q = (x,y)$. holonomic constraint $x^2+y^2 = l^2 \Rightarrow$ Pfaffian : $\underbrace{(x,y)}_{A}\cdot(\dot x, \dot y) = 0$.

$L(q,\dot q) = \frac{1}{2}m(\dot x^2 + \dot y^2) - mgy$, $\Upsilon = 0$.

$$\lambda = (AM^{-1}A^T)^{-1}\left[AM^{-1}(F - C\dot q - N) - \dot A \dot q\right)$$

$$= -\frac{m}{l^2}(gy + \dot x^2 + \dot y^2)$$

$$\Rightarrow \begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix}\begin{pmatrix} \ddot x \\ \ddot y \end{pmatrix} + \begin{pmatrix} 0 \\ mg \end{pmatrix} \underbrace{\underbrace{-\frac{m}{l^2}(gy + \dot x^2 + \dot y^2)}_{\lambda} \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{A}}_{\text{Tension of rod.}} = 0$$

$A = (x, y)$

$M = \begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix}$

$C = 0$

$N = \begin{bmatrix} 0 \\ mg \end{bmatrix}$

$F = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\dot A \dot q = \dot x^2 + \dot y^2$

<Lagrange - d'Alembert formulation>

In the previous example, we needed to first compute constraint force $A^T\lambda$ and then substitute to the motion of equation (Euler Lagrange equation).

Constraint force does no work to admissable direction

$\Rightarrow (A^T(q)\lambda)\cdot \delta q = 0 \quad \forall A(q)\delta q = 0$. $\quad \delta q$: virtual displacement.

So the equation of motion becomes

$$\left(\frac{d}{dt}\frac{\partial L}{\partial \dot q} - \frac{\partial L}{\partial q} - \Upsilon\right)\cdot \delta q = 0 \quad \forall A(q)\delta q = 0.$$

$\leftarrow$ Lagrange - d'Alembert equations.

Assume that $A(q)$ is full rank & $[A_1(q)\ A_2(q)]$ by reordering.
invertible

Then $A\delta q = 0 \Leftrightarrow \delta q_2 = -A_2^{-1}A_1\delta q_1, \quad \delta q_1 \in \mathbb{R}^{n-k}$ free.

Substitute to equation of motion:

$$\left(\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_1} - \frac{\partial L}{\partial q_1} - \Upsilon_1\right)\cdot \delta q_1 + \left(\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_2} - \frac{\partial L}{\partial q_2} - \Upsilon_2\right)\cdot(-A_2^{-1}A_1)\delta q_1 = 0.$$

$$\therefore \begin{cases} \dfrac{d}{dt}\dfrac{\partial L}{\partial \dot{q}_1} - \dfrac{\partial L}{\partial q_1} - \Upsilon_1 - A_1^T (A_2^T)^{-1}\left(\dfrac{d}{dt}\dfrac{\partial L}{\partial \dot{q}_2} - \dfrac{\partial L}{\partial q_2} - \Upsilon_2\right) = 0 \\ \dot{q}_2 = -A_2^{-1}A_1\dot{q}_1 \end{cases}$$

※ Nature of nonholonomic constraints

Suppose $\dot{q}_2 = -\tilde{A}(q_1)\dot{q}_1$. Then can we do Euler-Lagrange method to

$L(q_1, \dot{q}_1, -\tilde{A}(q_1)\dot{q}_1)$?

If we do it (assume no external force), we get

Lagrange - d'Alembert equation = Remainder term.

Right-hand side disappears if the constraint was holonomic.

6.2 Robot hand dynamics

<Derivation>

- Finger dynamics

$$M_f(\theta)\ddot{\theta} + C_f(\theta,\dot{\theta})\dot{\theta} + N_f(\theta,\dot{\theta}) = \tau.$$

- Object dynamics

$$\begin{bmatrix} mI & 0 \\ 0 & I \end{bmatrix}\begin{bmatrix} \dot{v}^b \\ \dot{\omega}^b \end{bmatrix} + \begin{bmatrix} \omega^b \times m v^b \\ \omega^b \times I\omega^b \end{bmatrix} = F^b$$

Use local parametrization of $SE(3) \ni (p, R)$, $x \in \mathbb{R}^6$ to get

$$M_o(x)\ddot{x} + C_o(x,\dot{x})\dot{x} + N_o(x,\dot{x}) = 0.$$

We combine two dynamics with velocity constraint (grasp constraint)

$J_h(\theta, x)\dot{\theta} = G^T(\theta, x)\dot{x}$ → velocity at contact point

Assume three things:

① force-closure, manipulable

↳ Finger can produce any wrench on the object

↳ Finger can produce any possible object motion (velocity) follow

② J_h is invertible

③ The contact force remains in friction cone; it does not slip or drop.

By ②, $A\begin{pmatrix}\theta \\ x\end{pmatrix}\begin{pmatrix}\dot{\theta} \\ \dot{x}\end{pmatrix} = [J_h \quad G^T]\begin{bmatrix}\dot{\theta} \\ \dot{x}\end{bmatrix}$ has invertible part.

So we can use d'Alembert's method. By $\delta\theta = J_h^{-1} G^T \delta x$, we get

$$\begin{cases} \left(\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x}\right) + G J_h^{-T}\left(\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta}\right) = G J_h^{-T}\tau & \text{n-m rows} \\ J_h\dot{\theta} = G^T\dot{x} & \text{m rows} \end{cases}$$

where $L = \frac{1}{2}\dot{\theta}^T M_f \dot{\theta} + \frac{1}{2}\dot{x}^T M_o \dot{x} - V_f(\theta) - V_o(x)$.

Substitute and eliminate $\dot{\theta}, \ddot{\theta}$ using the velocity constraint to get

$$\tilde{M}(q)\ddot{x} + \tilde{C}(q,\dot{q})\dot{x} + \tilde{N}(q,\dot{q}) = F.$$

It can be shown that ① $\tilde{M}$ symmetric, positive definite, ② $\dot{\tilde{M}} - 2\tilde{C}$ is skew-symmetric by properties of M_f, M_o, C_f, C_o.

<Internal force>

Internal force: contact force that has no net wrench on object.

e.g. $\rightarrow \square \leftarrow$

At stationary state, force at object is $G J_h^{-T} \tau$ (no gravity)

so we need $J_h^{-T} \tau \in N(G)$ to be internal.

But when moving, using the equation of motion in θ,

$M_f \ddot{\theta} + C_f \dot{\theta} + N_f - J_h^T \lambda = \tau$, contact force λ is

$\lambda = J_h^{-T}(\tau - M_f \ddot{\theta} - C_f \dot{\theta} - N_f)$.

And it is internal if $\lambda \in N(G)$

$\uparrow$

J_h^T converts force at contact to torque at joints.

6.3 Redundant, Nonmanipulable Robot System

Assumption ②, J_h being invertible can fail in two ways:

Redundant (1) Too many DOF: $N(J_h) \neq \{0\} \Leftrightarrow \operatorname{rank} J_h < n$.

Nonmanipulable (2) Too few DOF: $R(J_h) \neq \mathbb{R}^m \Leftrightarrow \operatorname{rank} J_h < m$.

$$\left(\begin{array}{ccc} J_h : \dot{\theta} \in \mathbb{R}^n & \mapsto & \dot{x}_c \in \mathbb{R}^m \\ \uparrow & & \uparrow \\ \text{joint velocity} & & \text{contact velocity} \end{array} \right)$$

(1) means there is an internal force

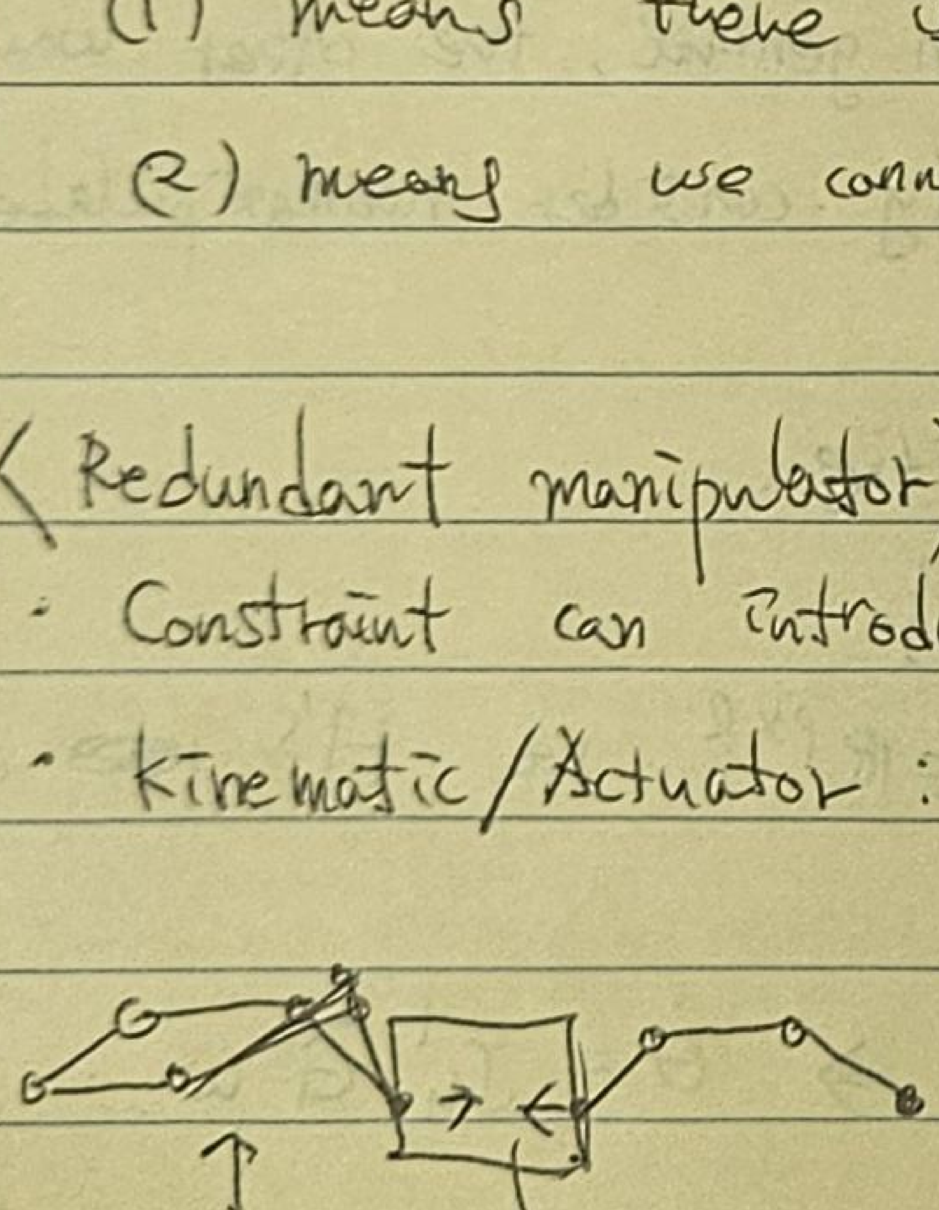
(2) means we cannot make some contact velocity.

<Redundant manipulator>

- Constraint can introduce DOF, e.g. soft-finger contact
- kinematic/Actuator : motion / force

$\downarrow$

internal force



$J_h \dot{\theta} = G^T \dot{x}$ null space of $\left(\begin{array}{l} J : \text{kinematic} \\ G^T : \text{Actuator} \end{array} \right)$

$\mathbb{R}^n$ $(p,R)\in$ Assume full row rank

Let $g:\theta \mapsto SE(3)$ be a kinematic map. If redundant,

$J_h \in \mathbb{R}^{6\times n}$ is not square. Let $K(\theta)$'s row spans null space of J_h.

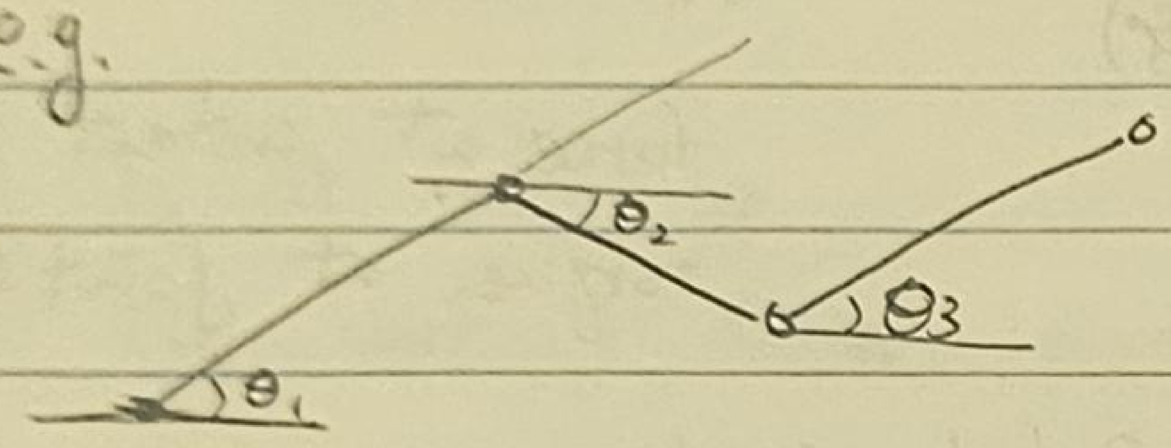
Define internal motion v_N by $\begin{pmatrix}\dot{x}\\ v_N\end{pmatrix} = \begin{pmatrix}J\\ K(\theta)\end{pmatrix}\dot{\theta} =: \bar{J}\dot{\theta}$.

Substituting to ~~motion~~ eq. of motion,

$$\tilde{M}(q)\begin{pmatrix}\ddot{x}\\ \dot{v}_N\end{pmatrix} + \tilde{C}(q,\dot{q})\begin{pmatrix}\dot{x}\\ v_N\end{pmatrix} + \tilde{N}(q,\dot{q}) = \bar{J}^{-T}\tau.$$

※ It would be nice if we can ~~extend~~ introduce $y=h(\theta)$ s.t. $v_N = \dot{y}$, i.e. $K = \frac{\partial h}{\partial \theta}$. This is the case when K is exact. ⇔ $\frac{\partial K_{ij}}{\partial \theta_k} = \frac{\partial K_{ik}}{\partial \theta_j}$.

e.g.



unit length

$x = \sum \cos\theta_i,\ y = \sum \sin\theta_i$

$$J(\theta) = \begin{pmatrix} -\sin\theta_1 & -\sin\theta_2 & -\sin\theta_3 \\ \cos\theta_1 & \cos\theta_2 & \cos\theta_3 \end{pmatrix}$$

~~$\theta)=\theta_3$ ⇐ If $\theta_1 \neq \theta_2$ ⇒ Take $k(\theta) = (0,0,1)$~~

$h(\theta)$ ⇐ If not $(\theta_1=\theta_2=\theta_3)$ ⇒ Take $K(\theta) = (\sin(\theta_2-\theta_1)\ \ -\sin(\theta_1-\theta_3)\ \ \sin(\theta_1-\theta_2))$

<Nonmanipulable grasp>

Now assume we have full column rank J. If not, we above technique.

Recall that manipulable is actually $R(G^T) \subseteq R(J_h)$.

So nonmanipulable implies not full row rank. In general, the other way is not true, but it is usually true. So we only consider nonmanipulable case.

Restrict the space by allowable object velocities:

$W(\theta, x) = \{\dot{x} \in \mathbb{R}^p \mid \exists \dot{\theta} \in \mathbb{R}^m \text{ s.t. } J_h \dot{\theta} = G^T \dot{x}\}$

Assume that $\dim W = l > 0$ const, $\exists H(\theta, x) \in \mathbb{R}^{p\times l}$ s.t. H's columns span W. Then $\exists w \in \mathbb{R}^l$ s.t.

$$\dot{x} = Hw \Rightarrow J_h\dot{\theta} = G^T H w = \bar{G} w. \Rightarrow \dot{\theta} = J_h^{+} \bar{G} w.$$

by pseudo-inverse

$$\Rightarrow \tilde{M}(q)\dot{w} + \tilde{C}(q,\dot{q})w + \tilde{N}(q,\dot{q}) = F.$$